

HARSHA DURVASULA

(412) 844-0581 | hdurvasu@andrew.cmu.edu | <https://www.linkedin.com/in/harsha-durvasula-1353971a4>

EDUCATION

Carnegie Mellon University, Pittsburgh, PA

Aug 2024 – Dec 2025

Master of Science, Information Systems

- **Coursework:** Cloud Computing, Engineering Data Intensive Scalable Systems, Data Structures, Distributed Systems, API Design
- **Teaching Assistantships:** Distributed Systems, Data Structures and Algorithms

PES University, Bangalore

Aug 2018 – May 2022

Bachelor of Technology, Computer Science Engineering

- **Coursework:** Database Management, Advanced Algorithms, Machine Learning, Artificial Intelligence, Graph Theory

SKILLS

Languages: Java, Python, C/C++, JavaScript, SQL

Cloud & DevOps: AWS (RDS, S3, ELB, EC2, ASG, Lambda, Cloud Formation), Azure, GCP, Kubernetes (EKS, AKS, GKE), Docker, Terraform, Helm, Jenkins, CI/CD, Grafana

Distributed Systems: Apache Kafka, Zookeeper, Apache Samza, Hadoop, Apache Spark, gRPC, REST, GraphQL, ETL pipelines

Databases: MySQL, PostgreSQL, MongoDB, Redis, RocksDB

Frameworks: Spring Boot, Maven, Django, Node.js, React, Vert.x

PROFESSIONAL EXPERIENCE

Carnegie Mellon University, Pittsburgh, PA

Jun 2025 – Present

Software Engineering Intern

- Developed a RAG-powered AI chatbot streamlining VR headset selection for **1,500+** CMU - Heinz students by auto-matching queries to relevant Meta Quest apps, reducing research time by **98%** (45min to 30sec) and increasing program utilization by **75%**
- Eliminated grading bottlenecks for **400+** student submissions by architecting an AI-powered auto-grading system using an LLM-based test generation and Docker, Kafka, Hadoop microservices, reducing evaluation time by **85%** while ensuring repeatable, precise results

Hevodata Inc, Bengaluru, India

Nov 2021 – May 2024

Software Engineer

- Accelerated webhook performance to **50K events/min** (2x improvement) for enterprise clients like Zepto by migrating Redis architecture to localized deployment, ensuring sub-second latency for time-sensitive ETL automation pipelines
- Processed **5M+** LinkedIn ad records daily across **500+** enterprise customers by building a distributed ETL pipeline with RabbitMQ coordination and Kafka streaming, achieving sub-minute data synchronization for business-critical analytics
- Cut security attack surface by **40%** by fortifying OAuth2.0 authentication and building comprehensive XSS prevention across user management APIs with **84** validation filters, safeguarding platform integrity for enterprise customers
- Elevated data pipeline efficiency by **30%** for **900+** enterprise customers through dynamic CSV header configuration in S3 connector, reducing manual data transformation steps and enabling immediate analytics on ingested datasets
- Streamlined data ingestion for **350+** customers by architecting compressed file handling (zip/rar/gz/tar.gz) directly into ETL pipelines across 3 cloud platforms, eliminating **60%** of manual data preprocessing and enabling automated multi-gigabyte workflows

PROJECTS

Book Recommender Microservice System

Jan 2025 – Mar 2025

- Scaled bookstore platform to **10K** concurrent users by architecting resilient **5-BFF** microservices Amazon EKS with ELB/ILB routing, AWS RDS MySQL persistence, and Kafka-SMTP pipeline automating customer onboarding emails
- Gained **3x** faster response times by implementing rate limiting, circuit breaker patterns, and Litmus chaos testing across 5 Java microservices, while automating deployments through CI/CD pipelines and async Kafka workflows

AWS Autoscaling and Elasticity [AWS EC2, ELB, CloudWatch]

Aug 2025 – Sept 2025

- Achieved 99% cluster uptime while reducing infrastructure costs by 40% by implementing Terraform-managed Auto Scaling Groups and Elastic Load Balancer with automated failover capabilities
- Refined scaling policies using CloudWatch metrics analysis and EC2 load simulation to achieve fault-tolerant, cost-efficient operation under variable demand

Kubernetes and Microservices [Google Kubernetes Engine, Azure Kubernetes Service]

Sept 2025 – Oct 2025

- Deployed fault-tolerant chat platform serving **5M+** daily messages across GCP/Azure Kubernetes with Helm automation, achieving **<200ms** latency, and zero-downtime deployments through multi-cloud orchestration
- Optimized performance by **70%** through polyglot database architecture (Redis for chat, MySQL for profiles, Neo4j/Cypher for social graphs, MongoDB for analytics) deployed via GitHub Actions CI/CD across GCP/Azure, achieving 80% faster deployment cycles

Kafka-Samza Real-Time Driver and Ad Matching Engine [AWS EMR, EC2, Apache Kafka, Apache Samza]

Nov 2025

- Engineered Kafka-Samza real-time ad engine processing **1.5GB+** event streams, matching riders to personalized venue advertisements in **<100ms** using health metrics and location-based scoring algorithms with **95%+** accuracy
- Increased distributed stream processing to **500K+** events using partitioned Kafka topics and Samza stateful processing with RocksDB stores, delivering personalized restaurant matches across **10K+** NYC venues with **40%** engagement lift